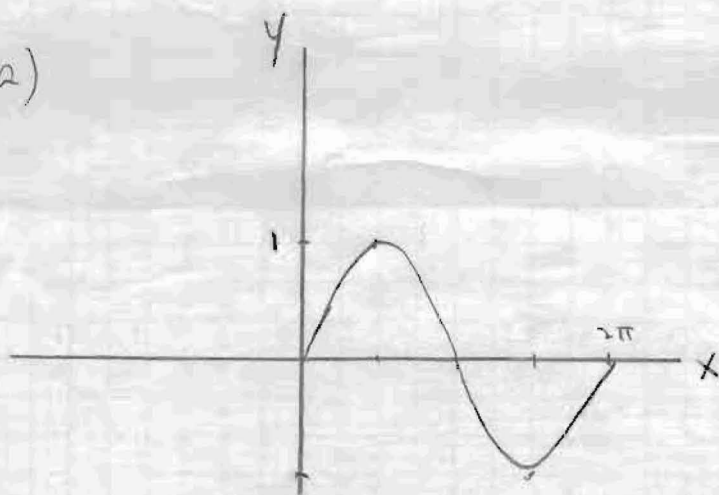
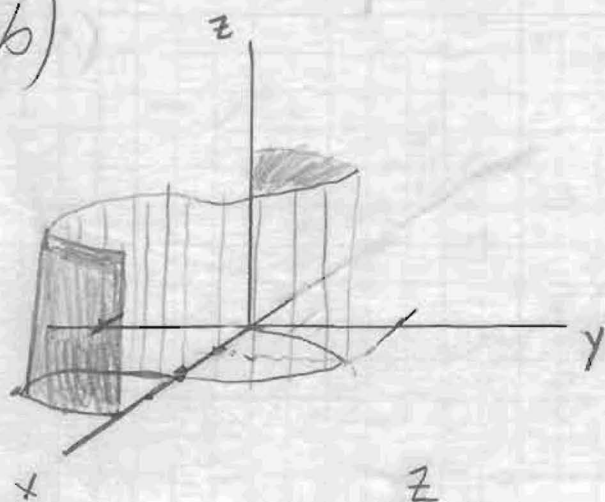


12.6

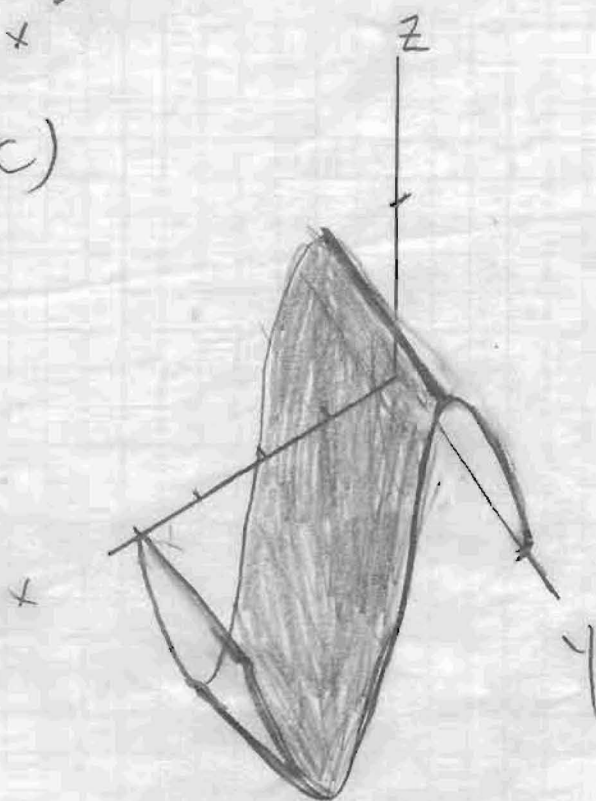
① a)



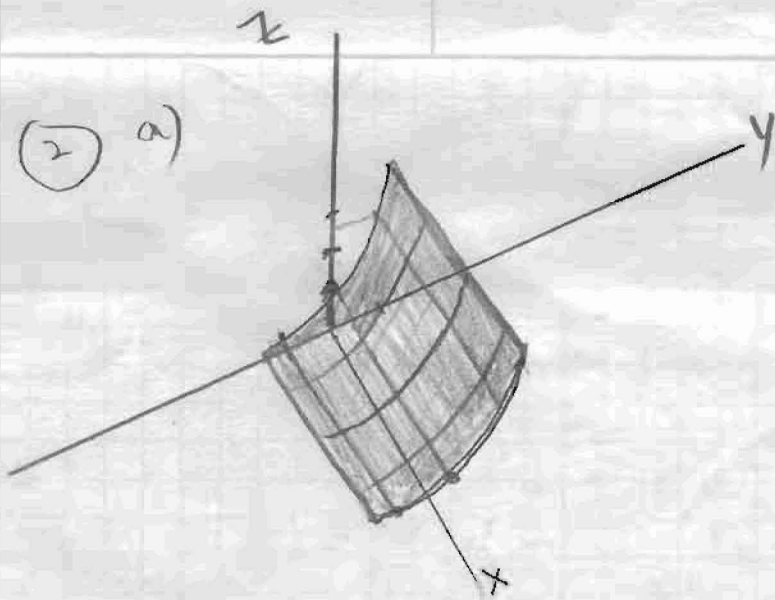
b)



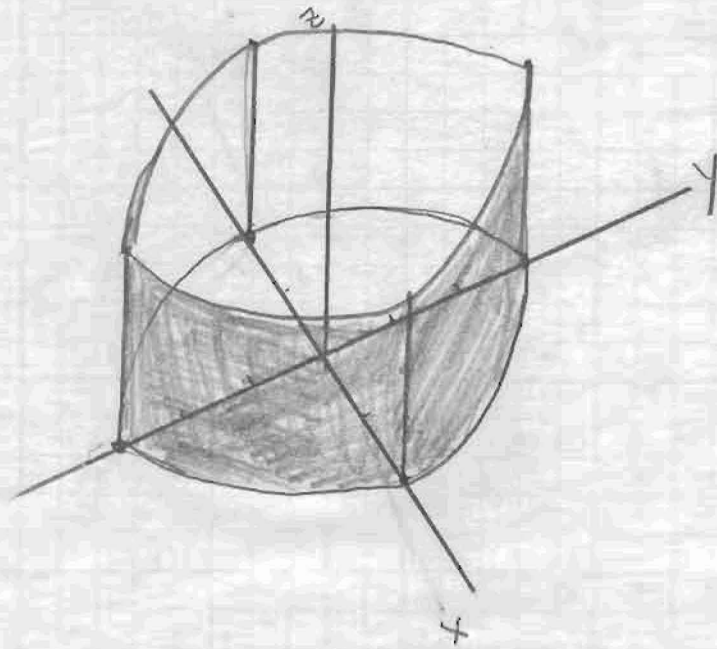
c)



(2) a)

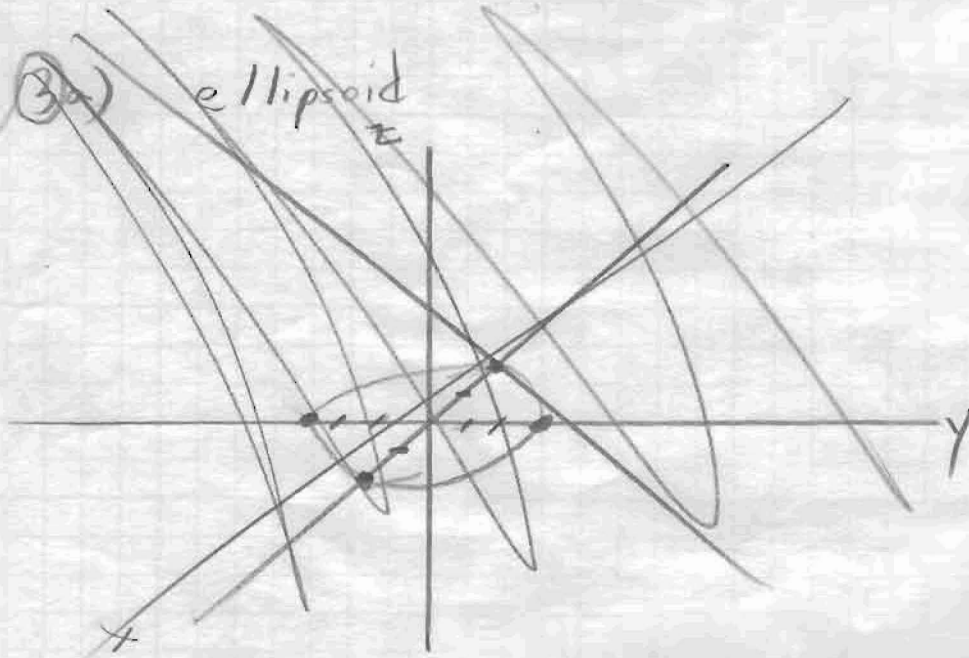


b)

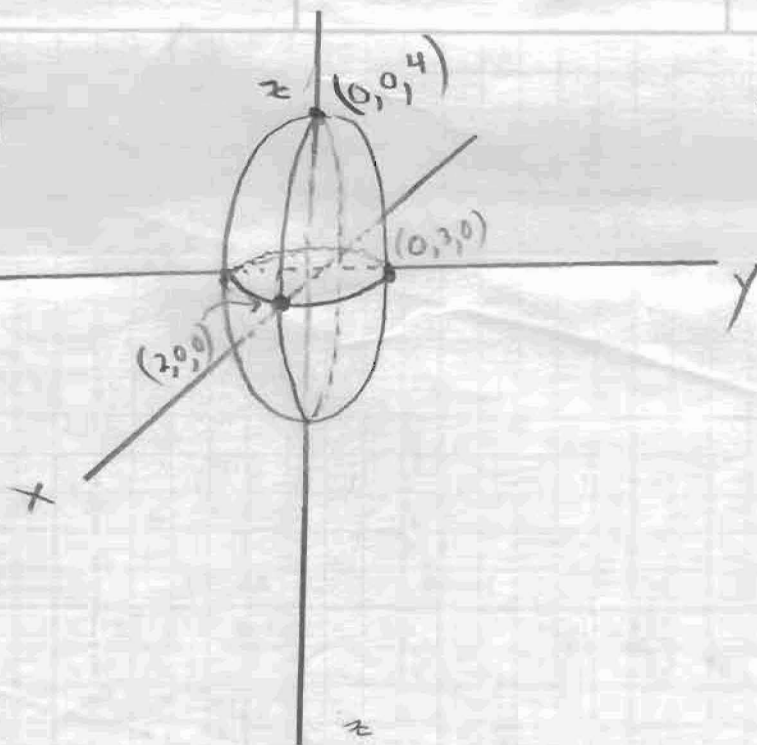


(3)

ellipsoid

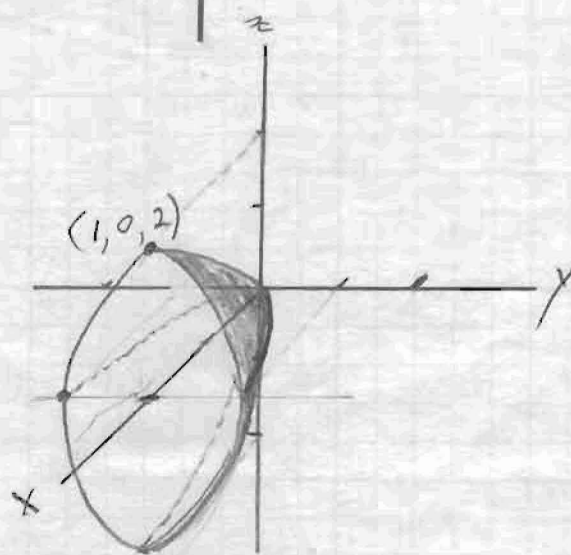


(3) a)



ellipsoid

b)

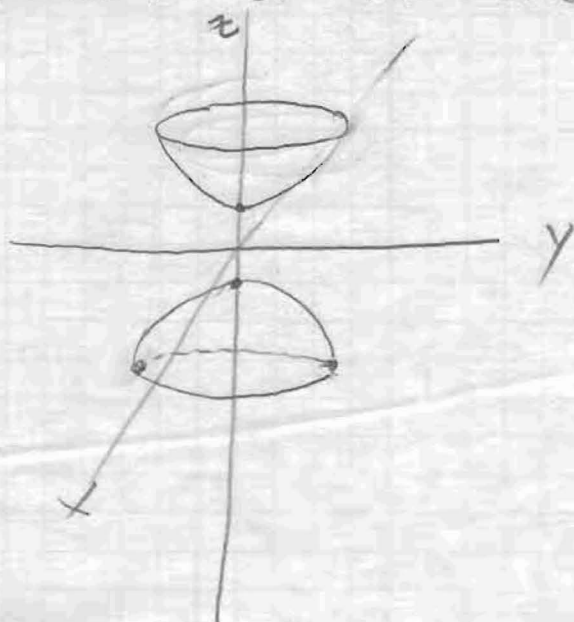


paraboloid

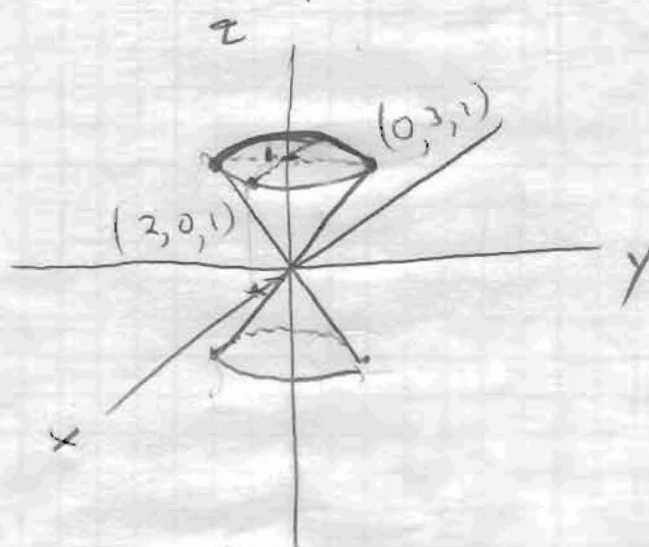
$$c) \quad z^2 = 1 + \frac{x^2}{4} + \frac{y^2}{9}$$

$$z^2 - \frac{x^2}{4} - \frac{y^2}{9} = 1$$

hyperboloid of two sheets

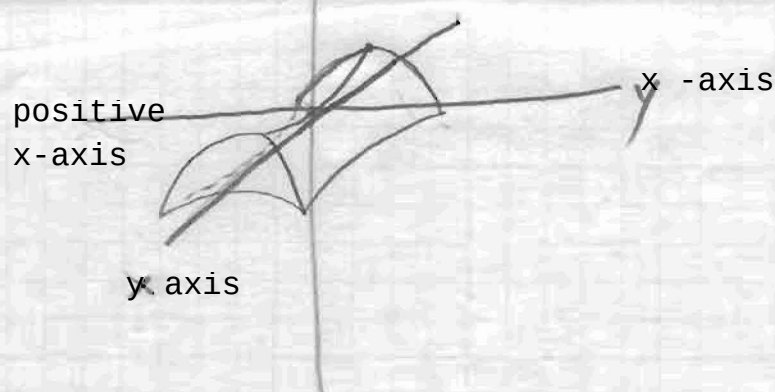


$$d) \quad z^2 = \frac{x^2}{4} + \frac{y^2}{9} \quad \text{cone}$$



e) $z = \frac{y^2}{16} - \frac{x^2}{9}$

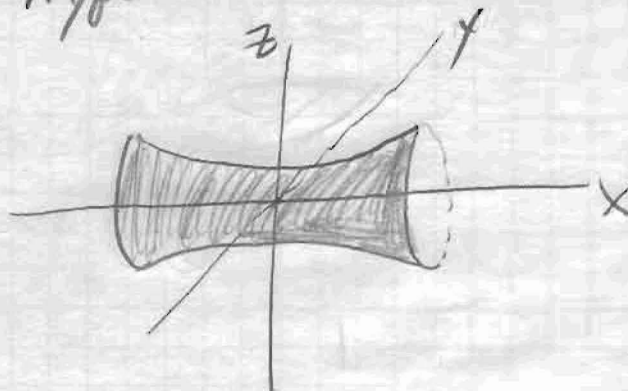
hyperbolic paraboloid



f)

$$y^2 + \frac{z^2}{9} - x^2 = 1$$

hyperboloid of one sheet



④

$$y^2 + 4y + 4x + z^2 - 10z + 29 = 0$$

$$y^2 + 4y + 4 + 4x + z^2 - 10z + 25 = 0$$

$$(y+2)^2 + 4x + (z-5)^2 = 0$$

$$4x = -[(y+2)^2 + (z-5)^2]$$

$$x = -\left[\frac{(y+2)^2}{4} + \frac{(z-5)^2}{4}\right]$$

Elliptic
Paraboloid

5) a) $\begin{matrix} 3 \times 2 & 2 \times 1 \\ \hline 3 \times 1 \end{matrix}$

b) $\begin{matrix} 2 \times 3 & 3 \times 1 \\ \hline 2 \times 1 \end{matrix}$

6) a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} -4 \\ 10 \end{bmatrix} = \begin{bmatrix} -4 + 20 \\ -12 + 40 \\ -20 + 60 \end{bmatrix} = \begin{bmatrix} 16 \\ 28 \\ 40 \end{bmatrix}$

b) $\begin{bmatrix} -5 & 0 & 3 \\ 2 & -4 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -11 + 0 + 9 \\ 4 + 4 + 18 \end{bmatrix} = \begin{bmatrix} -1 \\ 26 \end{bmatrix}$

7) $\mathbb{R}^3$ in $\mathbb{R}^2$

$\langle 2, -1, 3 \rangle$ to $\langle -1, 26 \rangle$